02 · Crime and deterrence: a county panel

Final report generated by the pipeline (python run_all.py). Tables and figures are the pipeline's own outputs; the text comes from the paper's README.

Question

Do counties with a higher probability of arrest, conviction and prison have less crime, and how much of the answer changes once fixed county traits, common year shocks and serial correlation are handled?

Result

| Specification | log P(arrest) | log police per capita | |---|---|---| | Pooled OLS | -0.633 | 0.352 | | County fixed effects | -0.395 | 0.422 | | Two-way fixed effects | -0.360 | 0.422 |

• Breusch–Pagan LM = 191.60 and White LM = 410.04, both with $p < 0.0001$: heteroskedasticity is clear.

• In the two-way model, clustering by county raises the standard error on the arrest elasticity from 0.033 to 0.064.

• On police per capita the clustered standard error is 0.093, against 0.027 before. Both coefficients keep three stars after clustering.

• Fixed effects cut the pooled arrest elasticity by about a third, so part of the pooled deterrence was county traits.

Reading: higher arrest probability goes with less crime even within a county. The positive police coefficient is the classic simultaneity sign (counties hire police where crime is high), so none of this is a causal estimate.

Data checks

10 pass, 2 warnings, 0 fail (details in output/data_checks.json).

Main tables

MQO agrupado vs. efeitos fixos de condado (within)

	MQO agrupado	EF de condado
	(1)	(2)
ln(prob. prisao) β_1	-0,633***	-0,395***
	(0,033)	(0,034)
ln(prob. condenacao)	-0,452***	-0,309***
	(0,022)	(0,022)
ln(prob. encarc.)	-0,150***	-0,211***
	(0,049)	(0,033)
ln(sentenca media)	-0,078**	0,032
	(0,038)	(0,026)
ln(policia p.c.) β_2	0,352***	0,422***
	(0,023)	(0,028)
Controles socioeconomicos	Sim	Absorvidos
Efeitos fixos de condado	Nao	Sim
Efeitos fixos de ano	Nao	Nao
Erros-padrao	Convencional	Convencional
N (obs.)	630	630
R^2	0,805	0,373

Source: output/q3_tab_fe_condado.tex

Efeitos fixos de condado vs. painel de duas vias (condado + ano)

	EF condado	EF duas vias
	(1)	(2)
ln(prob. prisao) β_1	-0,395***	-0,360***
	(0,034)	(0,033)
ln(prob. condenacao)	-0,309***	-0,286***
	(0,022)	(0,021)
ln(prob. encarc.)	-0,211***	-0,184***
	(0,033)	(0,033)
ln(sentenca media)	0,032	-0,003
	(0,026)	(0,027)
ln(policia p.c.) β_2	0,422***	0,422***
	(0,028)	(0,027)
Efeitos fixos de condado	Sim	Sim
Efeitos fixos de ano	Nao	Sim
Erros-padrao	Convencional	Convencional
N (obs.)	630	630
R^2	0,373	0,370

Source: output/q3_tab_duas_vias.tex

Painel de duas vias: erros convencionais vs. clusterizados por condado

	Convencional	Cluster
	(1)	(2)
ln(prob. prisao) β_1	-0,360***	-0,360***
	(0,033)	(0,064)
ln(prob. condenacao)	-0,286***	-0,286***
	(0,021)	(0,056)
ln(prob. encarc.)	-0,184***	-0,184***
	(0,033)	(0,049)
ln(sentenca media)	-0,003	-0,003
	(0,027)	(0,037)
ln(policia p.c.) β_2	0,422***	0,422***
	(0,027)	(0,093)
Efeitos fixos de condado	Sim	Sim
Efeitos fixos de ano	Sim	Sim
Erros-padrao	Convencional	Cluster (condado)
N (obs.)	630	630
R^2	0,370	0,370

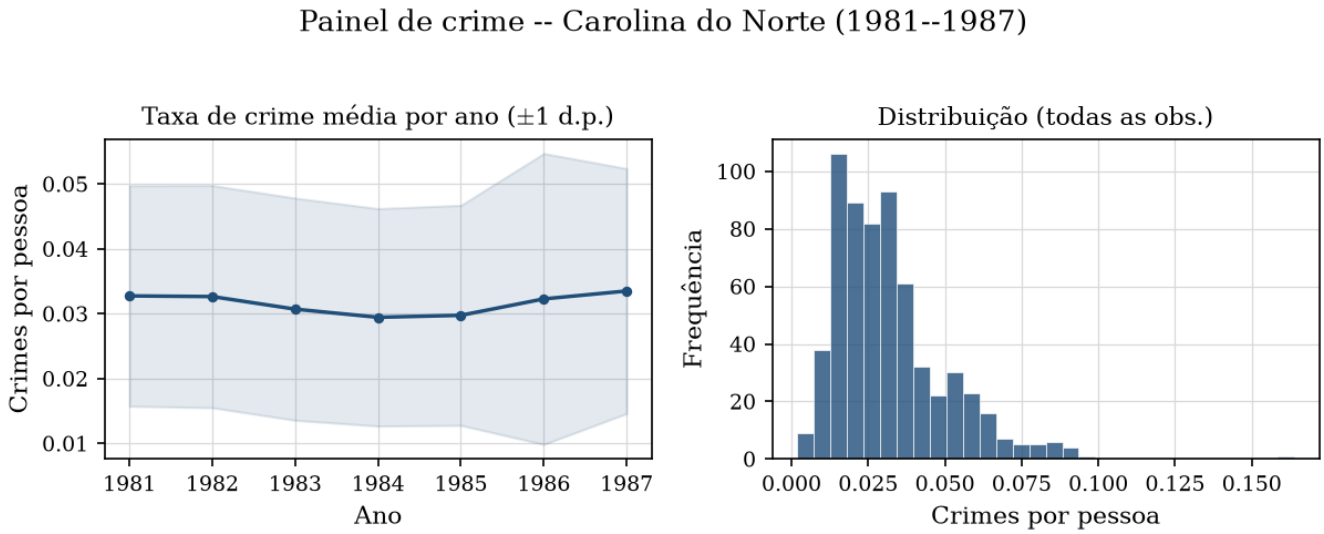
Source: output/q3_tab_cluster.tex

Testes de heterocedasticidade no MQO agrupado

Teste	Estatistica LM	g.l.	p-valor
Breusch–Pagan	191,60	14	<0,0001
White	410,04	115	<0,0001

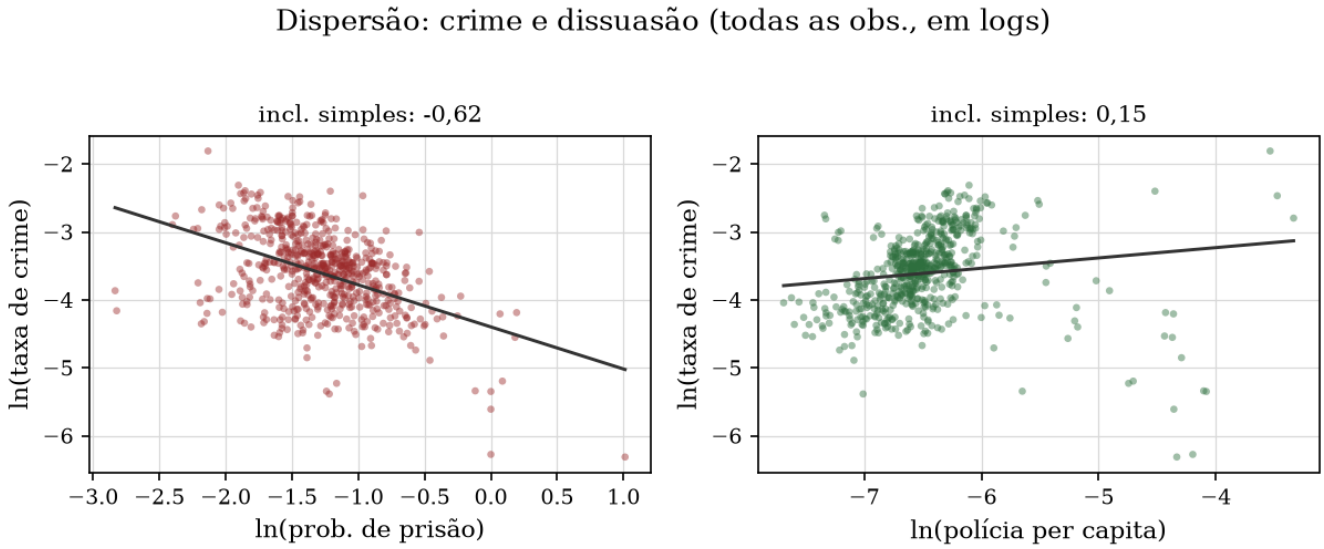
Source: output/q3_tab_heterocedasticidade.tex

Main figures



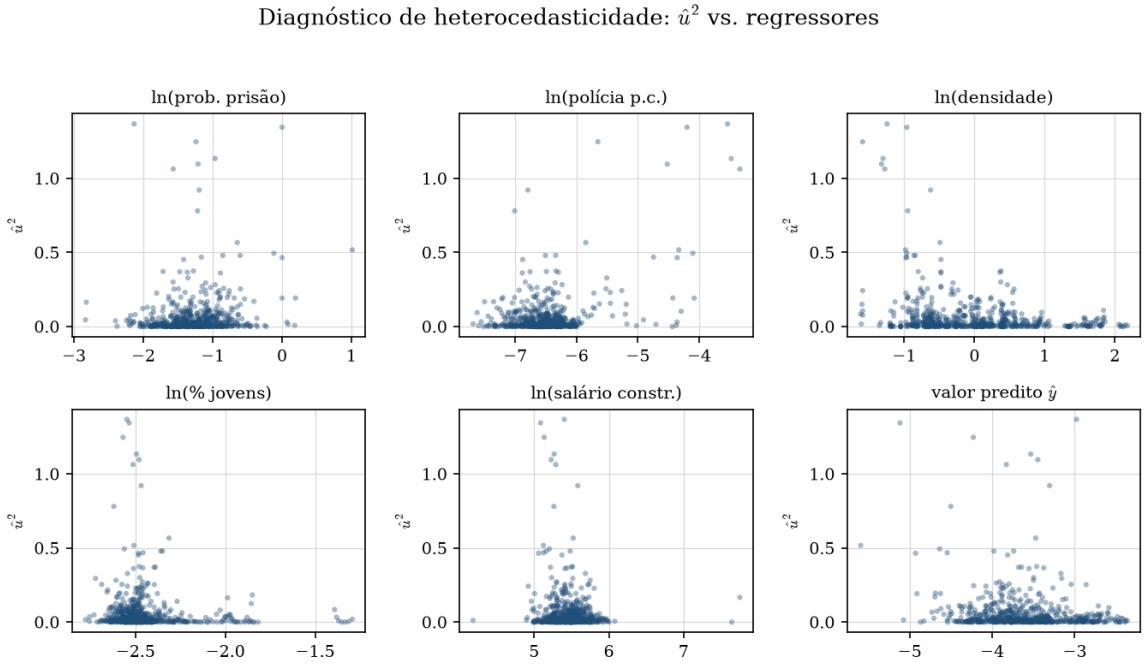
Fonte: Cornwell e Trumbull (1994), painel de 90 condados da Carolina do Norte (1981-1987), via wooldridge::crime4. Elaboracao propria.

Source: output/q3_fig_painel_descritivo.png



Fonte: Cornwell e Trumbull (1994), painel de 90 condados da Carolina do Norte (1981-1987), via wooldridge::crime4. Elaboracao propria.

Source: output/q3_fig_dispersao.png



Fonte: Cornwell e Trumbull (1994), painel de 90 condados da Carolina do Norte (1981-1987), via wooldridge::crime4. Elaboracao propria.

Source: output/q3_fig_residuos_scatter.png